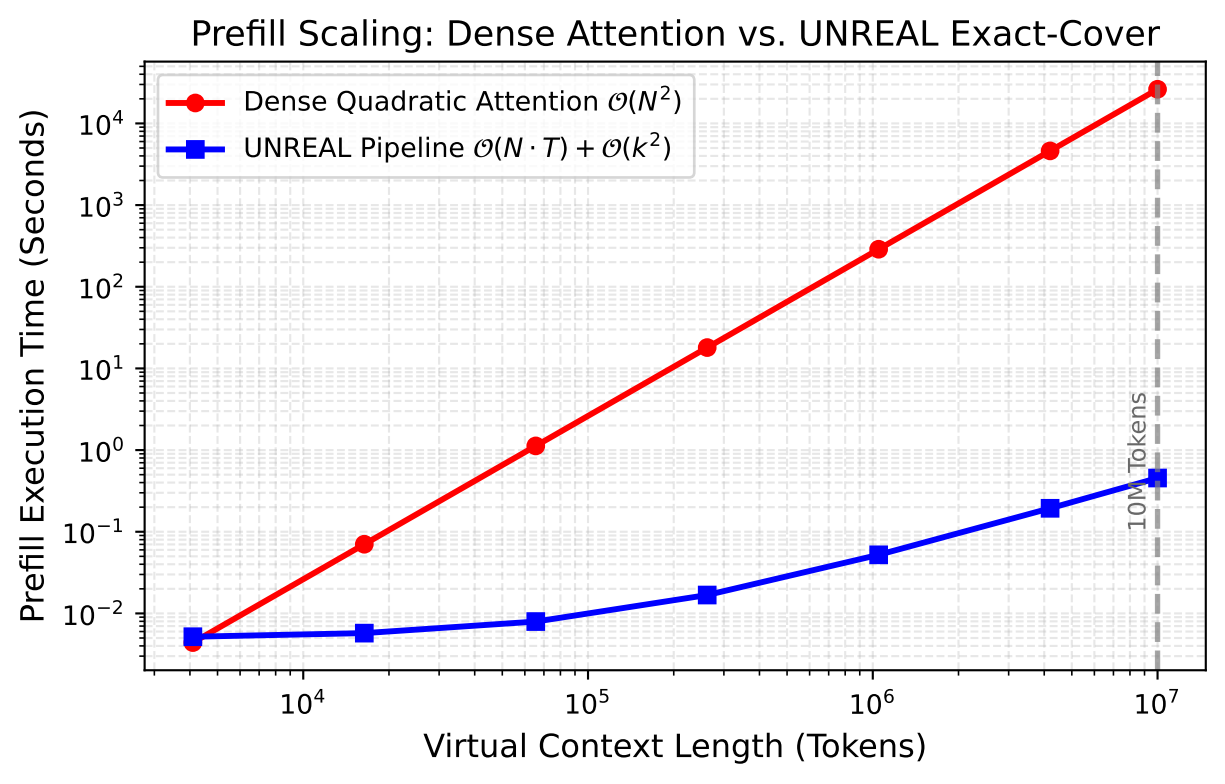
Prefill Scaling: Dense Attention vs. UNREAL Exact-Cover
Dense Quadratic Attention $\mathcal{O}(N^2)$
UNREAL Pipeline $\mathcal{O}(N \cdot T) + \mathcal{O}(k^2)$
Prefill Execution Time (Seconds)
Virtual Context Length (Tokens)
10M Tokens
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{4}
10^{5}
10^{6}
10^{7}